

RELIABLE RENTALS

Conceptual Database Design Report

Elkin Huertas, Yasemin Yilmaz, Jennifer Rogers

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1. Introduction

This report presents the conceptual database design for Reliable Rentals, a vehicle hire company that operates across multiple outlet locations. The purpose of this document is to systematically develop and document a conceptual data model that accurately reflects the business's informational requirements.

The conceptual model is developed in accordance with standard Entity-Relationship (ER) modelling methodology.

Scope

The model encompasses the following operational areas of Reliable Rentals:

- Vehicle inventory management across outlet locations
- Client registration and profile management
- Hire agreement processing and tracking
- Staff management across all outlets

Methodology

The conceptual data model is produced using the Entity-Relationship (ER) approach, following these steps in sequence:

- Identification of main entities from the case study requirements
- Identification of relationships between entities
- Determination of multiplicity constraints for each relationship
- Identification and assignment of attributes to entities
- Determination of candidate and primary keys for each entity
- Generation of the ER diagram at the conceptual level (no foreign keys as attributes)

2. Design Assumptions

The following assumptions have been made during the conceptual design process. These assumptions address ambiguities in the case study.

#	Assumption
A1	A vehicle is located at exactly one outlet at any given point in time. Vehicles may be relocated between outlets, but the current location is a single outlet.
A2	A hire agreement is associated with exactly one vehicle and exactly one client. A single hire agreement cannot cover multiple vehicles or multiple clients.
A3	Each client must have at least one hire agreement on file in the system. However, the model is designed so that a client can exist with zero hire agreements (e.g., a registered client who has not yet hired a vehicle).
A4	Each vehicle must have been involved in at least one hire agreement for history to be recorded, but vehicles may exist in the system with zero hire agreements (newly acquired vehicles not yet hired).
A5	Each outlet must have at least zero vehicles and zero staff at any given time (e.g., a new outlet being set up). There is no enforced minimum.
A6	Client Number is selected as the primary key for the CLIENT entity rather than the Driving License Number. The Driving License Number is treated as an alternate (candidate) key.
A7	The redundant client and vehicle data stored in the original hire agreement description (client name, address, phone, vehicle model, make) is not modelled as attributes of the HIRE AGREEMENT entity. This data is derivable through relationships, and its inclusion would violate normalization principles.
A8	Each staff member is assigned to exactly one outlet.
A9	Name attributes (for both clients and staff) are stored as two separate attributes: FirstName and LastName. A full name may be derived by concatenation.
A10	Address attributes (for outlets, clients, and staff) are treated as a single composite string attribute at the conceptual level.

3. Step A — Identification of Main Entities

Entities represent the principal objects or concepts that the business needs to store information about. The following five entities were identified from the Reliable Rentals case study. Each entity corresponds to a distinct, clearly identifiable object type with multiple instances and independently stored data.

Entity List

OUTLET

Represents a physical location (branch) operated by Reliable Rentals where vehicles are stocked and staff are based. Each outlet is uniquely identifiable and has its own contact details.

VEHICLE

Represents a vehicle available for hire in the company fleet. Each vehicle is uniquely identified by its registration number and is associated with a specific outlet as its current location.

CLIENT

Represents an individual who registers with Reliable Rentals to hire vehicles. Clients have personal and contact details stored for identification and communication purposes.

HIRE AGREEMENT

Represents a formal contractual agreement between a client and Reliable Rentals for the hire of a specific vehicle over a defined period. Each hire agreement is a distinct transaction uniquely identifiable by its hire number.

STAFF

Represents an individual employed by Reliable Rentals and based at a specific outlet. Staff records include personal details, employment history, and role information.

4. Step B — Identification of Relationships

A relationship represents a meaningful business association between two or more entities. The following relationships were identified based on the operational interactions described in the case study.

Relationship	Entities	Multiplicity	Description
Stocks	OUTLET — VEHICLE	1 to 0..*	An outlet stocks a collection of vehicles. Each vehicle is stocked at exactly one outlet.
Employs	OUTLET — STAFF	1 to 0..*	An outlet employs members of staff. Each staff member is employed at exactly one outlet.
Makes	CLIENT — HIRE AGREEMENT	1 to 1..*	A client makes one or more hire agreements. Each hire agreement is made by exactly one client.
Covered By	VEHICLE — HIRE AGREEMENT	1 to 1..*	A hire agreement covers exactly one vehicle. Each vehicle may appear in one or more hire agreements over time.

5. Step C — Multiplicity Constraints

OUTLET — VEHICLE (Stocks)

- OUTLET side (1..1): Each vehicle must be associated with exactly one outlet. A vehicle cannot exist in the system without a current location.
- VEHICLE side (0..*): An outlet may have zero or more vehicles stocked. A new outlet may temporarily have no vehicles allocated to it.

OUTLET — STAFF (Employs)

- OUTLET side (1..1): Each staff member is based at exactly one outlet. A staff member cannot be unassigned.
- STAFF side (0..*): An outlet may employ zero or more staff members. An outlet being set up may temporarily have no staff assigned.

CLIENT — HIRE AGREEMENT (Makes)

- CLIENT side (1..1): Each hire agreement is made by exactly one client. A hire agreement cannot exist without a client.
- HIRE AGREEMENT side (0..*): A client may have zero or more hire agreements. A newly registered client may not yet have made a hire.

VEHICLE — HIRE AGREEMENT (Covered By)

- VEHICLE side (1..1): Each hire agreement covers exactly one vehicle. A hire agreement must reference a specific vehicle.
- HIRE AGREEMENT side (0..*): A vehicle may appear in zero or more hire agreements. A vehicle newly added to the fleet may not yet have been hired.

6. Step D — Attributes

OUTLET

Attribute	Data Type	Key	Description
OutletNumber	<i>String</i>	PK	Unique identifier assigned to each outlet
Address	<i>String</i>		Full street address of the outlet
PhoneNumber	<i>String</i>		Contact phone number for the outlet
FaxNumber	<i>String</i>		Fax number for the outlet

VEHICLE

Attribute	Data Type	Key	Description
RegistrationNumber	<i>String</i>	PK	Unique vehicle registration plate number
Model	<i>String</i>		Vehicle model name
Make	<i>String</i>		Vehicle manufacturer/brand
EngineSize	<i>String</i>		Engine displacement
Capacity	<i>Integer</i>		Passenger seating capacity
CurrentMileage	<i>Integer</i>		Odometer reading at time of last update
DailyHireRate	<i>Decimal</i>		Cost per day to hire the vehicle

CLIENT

Attribute	Data Type	Key	Description
ClientNumber	<i>String</i>	PK	Unique system-generated client identifier
FirstName	<i>String</i>		Client's first/given name
LastName	<i>String</i>		Client's surname
HomeAddress	<i>String</i>		Client's residential address
PhoneNumber	<i>String</i>		Client's contact telephone number
DateOfBirth	<i>Date</i>		Client's date of birth
DrivingLicenseNumber	<i>String</i>	CK	Government-issued driving license number (alternate key)

HIRE AGREEMENT

Attribute	Data Type	Key	Description
HireNumber	<i>String</i>	PK	Unique identifier for each hire agreement
StartDate	<i>Date</i>		Date on which the hire period commences
EndDate	<i>Date</i>		Date on which the client intends to return the vehicle
MileageBefore	<i>Integer</i>		Vehicle odometer reading at start of hire
MileageAfter	<i>Integer</i>		Vehicle odometer reading at end of hire

STAFF

Attribute	Data Type	Key	Description
StaffNumber	<i>String</i>	PK	Unique identifier assigned to each staff member
FirstName	<i>String</i>		Staff member's first/given name
LastName	<i>String</i>		Staff member's surname
HomeAddress	<i>String</i>		Staff member's residential address
HomePhoneNumber	<i>String</i>		Staff member's personal contact number
DateOfBirth	<i>Date</i>		Staff member's date of birth
Sex	<i>String</i>		Staff member's sex/gender as recorded
DateJoined	<i>Date</i>		Date the staff member joined the company
JobTitle	<i>String</i>		Staff member's job role or title
Salary	<i>Decimal</i>		Staff member's annual salary

7. Step E — Candidate and Primary Keys

Entity	Candidate Key(s)	Primary Key	Rationale
OUTLET	OutletNumber	OutletNumber	Single candidate key
VEHICLE	RegistrationNumber	RegistrationNumber	Single candidate key
CLIENT	ClientNumber DrivingLicenseNumber	ClientNumber	Two candidates; ClientNumber selected as stable key
HIRE AGREEMENT	HireNumber	HireNumber	Single candidate key
STAFF	StaffNumber	StaffNumber	Single candidate key

8. Step F — Conceptual ER Diagram

The Entity-Relationship diagram below presents the complete conceptual data model for Reliable Rentals.

Reliable Rentals - Conceptual ER Diagram

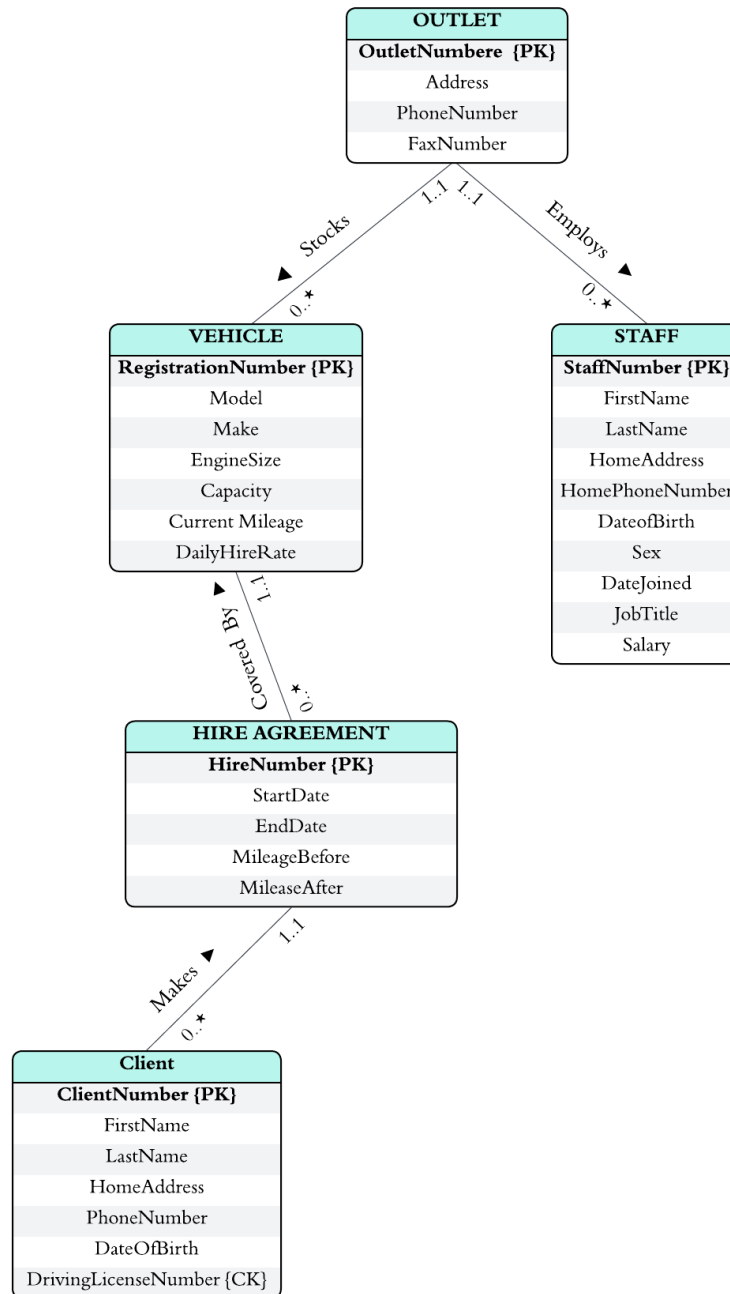


Diagram Description

The diagram contains five entities arranged to clearly show the four key relationships:

- OUTLET stocks VEHICLE (1..1 to 0..*): an outlet holds zero or many vehicles; each vehicle belongs to exactly one outlet.
- OUTLET employs STAFF (1..1 to 0..*): an outlet employs zero or many staff; each staff member belongs to exactly one outlet.
- CLIENT makes HIRE AGREEMENT (1..1 to 0..*): a client makes zero or many hire agreements; each agreement belongs to exactly one client.
- VEHICLE covered by HIRE AGREEMENT (1..1 to 0..*): a vehicle appears in zero or many hire agreements; each agreement covers exactly one vehicle.